

# HTML — Complete Learning Reference

**Prepared for:** Indana Rishi Yadztabika **Purpose:** A complete, unfiltered list of HTML topics for self-directed review. Mark what's shaky and study accordingly — nothing here is pre-ranked by importance.

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## 1. Document & Syntax Fundamentals

Doctype, HTML living standard vs "HTML5" versioning

Document tree, elements vs tags vs nodes

Attributes: global attributes, boolean attributes, data-\* attributes, custom attributes

Void/self-closing elements

Comments

Character encoding (`< charset >`), entity references (`&nbsp;`, `&`, etc.)

Whitespace handling rules

Case sensitivity conventions

`< head >` contents: `< title >`, `< base >`, `< meta >` (all variants), `< link >` (all `rel` types), `< style >`, `< script >`

placement/attributes (`async`, `defer`, `type=module`)

## 2. Sectioning & Structure

`< html >`, `< body >`

`< header >`, `< footer >`, `< main >`, `< nav >`, `< article >`, `< section >`, `< aside >`

`< div >` and when it's actually correct

Heading elements `< h1 >`–`< h6 >`, `< hgroup >`

Outline algorithm (historical vs current spec status)

## 3. Text Content

`< p >`, `< blockquote >`, `< pre >`, `< hr >`

Lists: `< ul >`, `< ol >`, `< li >`, `< dl >`/`< dt >`/`< dd >`

Inline text semantics: `< a >`, `< em >`, `< strong >`, `< small >`, `< s >`, `< cite >`, `< q >`, `< dfn >`, `< abbr >`,

`< ruby >`/`< rt >`/`< rp >`, `< data >`, `< time >`, `< code >`, `< var >`, `< samp >`, `< kbd >`, `< sub >`/`< sup >`, `< i >`, `< b >`, `< u >`,

`< mark >`, `< bdi >`/`< bdo >`, `< span >`, `< br >`, `< wbr >`

## 4. Links & Navigation

`< a >` — `href`, `target`, `rel` (all values, including security-relevant ones), `download`, `ping`

URL schemes (`mailto:`, `tel:`, `javascript:` and why to avoid it)

Fragment identifiers and same-page navigation

## 5. Embedded Content

`< img >` — `src`, `srcset`, `sizes`, `alt`, `loading`, `decoding`, `fetchpriority`

`< picture >`, `< source >`

`<audio>`, `<video>`, `<track>`

`<iframe>` — `sandbox`, `allow`, `loading`, `referrerpolicy`

`<embed>`, `<object>`, `<param>`

`<map>`, `<area>` (image maps)

`<canvas>`

SVG embedding (inline, `<img>`, `<object>`)

## 6. Tables

`<table>`, `<caption>`, `<colgroup>`/`<col>`, `<thead>`/`<tbody>`/`<tfoot>`, `<tr>`, `<th>`, `<td>`

`scope`, `headers`, `id` attributes for accessible tables

`rowspan`/`colspan`

## 7. Forms — Full Surface

`<form>` — `action`, `method`, `enctype`, `target`, `novalidate`, `autocomplete`

`<input>` — every `type` value (text, password, email, tel, url, number, range, date, time, datetime-local, month, week, color, checkbox, radio, file, hidden, search, submit, reset, button, image)

`<textarea>`, `<select>`/`<option>`/`<optgroup>`, `<datalist>`

`<button>` (all `type` values)

`<label>`, `<fieldset>`, `<legend>`, `<output>`, `<progress>`, `<meter>`

Validation attributes: `required`, `pattern`, `min`, `max`, `step`, `minlength`, `maxlength`

Constraint Validation API, `ValidityState`

`FormData` API

Autofill/`autocomplete` token values

## 8. Interactive Elements

`<details>`/`<summary>`

`<dialog>` — `open`, `showModal()`, `close()`, `:::backdrop`

Popover API — `popover` attribute, `popovertarget`, `popovertargetaction`

Invoker Commands — `command`, `commandfor` attributes

`<template>`

## 9. Web Components (native, framework-independent)

Custom Elements

Shadow DOM, Declarative Shadow DOM

HTML Templates

Slots

## 10. Accessibility (ARIA + native semantics)

ARIA roles (landmark, widget, document structure, live region roles)

ARIA states/properties (`aria-expanded`, `aria-hidden`, `aria-live`, `aria-labelledby`, `aria-describedby`, etc.)

Implicit ARIA semantics of native elements

`tabindex` (0, -1, positive values and why to avoid them)

Focus management APIs

`alt`, `title` attribute proper usage vs misuse

## 11. Metadata, SEO & Social

`<meta name="description">`, `<meta name="viewport">`, `<meta charset>`, `<meta http-equiv>`

Open Graph tags, Twitter Card tags

`<link rel="canonical">`, `preload`, `prefetch`, `preconnect`, `dns-prefetch`, `modulepreload`

Favicon/manifest links

JSON-LD structured data

robots meta directives

## 12. Security-Relevant HTML

`rel="noopener noreferrer"`

`<iframe sandbox>` values

Content Security Policy interaction with inline `<script>`/`<style>`

`integrity`/`crossorigin` attributes (Subresource Integrity)

`X-Frame-Options` relevance to `<iframe>` embedding

## 13. Internationalization

`lang` attribute (document and element-level)

`dir` attribute, `<bdi>`/`<bdo>`

Character encoding edge cases

## 14. Performance-Relevant HTML

`loading="lazy"`, `fetchpriority`

Resource hints (`preload`, `prefetch`, `preconnect`)

`async`/`defer` on `<script>`

Critical rendering path basics (how HTML parsing blocks/unblocks on CSS/JS)

## 15. Deprecated/Historical (worth knowing to recognize legacy code)

`<center>`, `<font>`, `<marquee>`, `<blink>`, frame-based layouts (`<frameset>`, `<frame>`)

Presentational attributes (`bgcolor`, `align`, etc.) and why they were replaced by CSS

## 16. Validation & Tooling

HTML validators (W3C validator)

Linting (HTMLHint or equivalent)

Browser DevTools: Elements panel, Accessibility tree inspector

## How to Use This Document

Go through each section and mark topics as: **Solid / Rusty / New to me**

Prioritize your own study order — this list is intentionally unranked

For "New to me" items, don't just read — rebuild something real using them

Suggested verification exercise once you've reviewed a section: rebuild a real page from your portfolio using zero unnecessary `<div>`s, run a Lighthouse accessibility audit, and navigate it using only a keyboard

*This is Topic 1 of an ongoing self-study series covering HTML, CSS, TypeScript, Testing, Accessibility, Performance, Angular basics, and Node.js/BFF patterns — tailored for frontend developer roles in Germany, Netherlands, Malaysia, and UAE.*